

Mind Space — Product Requirements Document

Document 01 of 06 | Version 1.0 | PFE Project

App overview

App Name:

Mind Space

Tagline:

Your emotional memory. Not a journal. Not a chatbot. A mirror that remembers.

Platform:

Android (Flutter) — v1. iOS stretch goal.

Problem statement

Most people carry the same emotional themes across weeks and months — a difficult job search, a family conflict, recurring self-doubt. Current journaling apps treat each entry as a standalone event. AI chatbots forget everything the moment the conversation ends. There is no tool that automatically notices when today's conversation is part of a longer-running emotional theme, groups those conversations together, and eventually reflects back how the user has been changing over time.

Mind Space solves this by automatically identifying recurring themes (called Arcs) across conversations, and generating macro-level insights once a theme has enough sessions to reveal a pattern.

Target user

Primary persona:

- Age 18–35, university students and young professionals
- Introspective — already does some form of journaling or self-reflection
- Not seeking therapy, but wants more than a generic AI chat experience
- Comfortable with mobile apps; privacy-conscious about emotional data

Core value proposition

Mind Space does three things no other app does together:

1. Automatic theme detection — the user just talks; Arcs form themselves via embedding-based clustering
2. Longitudinal emotional memory — Sage's responses in later sessions are informed by the Arc's history
3. Macro insight generation — after 3+ sessions, users can see how they have been changing, not just what they said

Feature list

Must have (v1)

Feature	Description	Priority
Sage chat (streaming)	Conversational AI with Person-Centered + EFT-informed prompt. Streaming response token by token.	P0
Post-chat reflection	Auto-generated reflection card: emotion spirit, what Sage heard (EFT-layered), one question to sit with.	P0
Arc auto-assignment	Embedding-based clustering assigns each reflection to an existing Arc or creates a new one (cosine similarity threshold 0.78).	P0
Arc detail view	Session list, emotional journey snake, generate insight button (unlocks at ≥3 sessions).	P0
Arc insight generation	Macro LLM analysis across all sessions in an Arc. Structured as: how it evolved + pattern noticed.	P0
Timeline history	Reverse-chronological list of all reflections, filterable by emotion.	P1
Arcs grid	2-column card grid of all Arcs with emotion dot rows.	P1
Auth (Google + magic link)	No passwords. Google OAuth or email magic link only.	P0
Safety classifier	Pre-chat message check. Binary SAFE/CRISIS. Crisis triggers pre-written response + resources card.	P0
Cross-arc context injection	Reflection prompt includes last what_sage_heard from other active Arcs for whole-person awareness.	P1
EFT processing stage tracking	Each Arc has a processing_stage field (forming → venting → stabilizing → processing → shifting → integrating). Sage behavior tuned per stage.	P1
Settings (minimal)	Sign out, export data as JSON, delete account.	P1

Nice to have (v2)

- Voice input (mic button in chat)
- Arc merging suggestion when two Arc centroids drift within 0.85 similarity
- Cross-arc case formulation view (predisposing / precipitating / perpetuating / protective factors)
- Light mode
- iOS build
- Push notifications for check-ins

Out of scope — v1

- Clinical diagnosis or any claim of therapeutic efficacy
- Human therapist integration or escalation paths
- Social features (sharing, community)
- Web app version
- Paid subscription / monetization layer
- Multi-language UI (app defaults to English; crisis resources localized to DZ)

User stories

As a...	I want to...	So that...
new user	open the app and immediately start a conversation without a signup form blocking me	I experience the product before committing
returning user	see my open Arcs on the home screen with a context-aware opener from Sage	I can pick up exactly where I left off
user after a chat	see a reflection card that names what I was actually feeling underneath the surface	I feel genuinely understood, not just logged
user with 3+ sessions in an Arc	generate a macro insight that shows me how I have been changing	I can see my own growth that I could not see from inside it
user in distress	see calm, appropriate crisis resources without the app panicking	I feel held, not alarmed
user re-reading history	open any old reflection and see it exactly as it was	I can track my emotional evolution over time
user who disagrees with an Arc assignment	move a session to a different Arc or create a new one	the Arc system feels collaborative, not imposed

Success metrics

Metric	Target	Measurement
Reflection quality score	$\geq 7.5 / 10$	Manual eval on 30 test chats across 5 dimensions (warmth, accuracy, non-prescriptiveness, brevity, no-overreach)
Arc coherence score	$\geq 85\%$ correct assignments	Manual inspection of 20 diverse reflection pairs vs. arc assignments
Safety classifier precision	0 false negatives	Manual eval on 15 crisis-signal test messages
Safety classifier recall	$< 10\%$ false positive rate on distressed-but-safe messages	Manual eval on 15 distressed-but-safe test messages
Demo flow completion	5-min demo runs end-to-end without error	Tested on 3 Android devices
Cold start time	< 2 seconds to home screen	Manual timing on mid-range Android device